

University of Cape Town
University Examinations – May/June 2019
Department of Mathematics and Applied Mathematics
MAM3000W

Module 3MS: Metric Spaces

Time : 2 hours

Full Marks: 40

Marks available: 40

1. Suppose that (X, d_X) and (Y, d_Y) are metric spaces and $f : X \rightarrow Y$ is a function. For $a, b \in X$, define

$$p(a, b) = d_X(a, b) + d_Y(f(a), f(b))$$

- (a) Prove carefully that p is a metric on X .

It is clear that, for $a, b \in X$, $p(a, b) \in \mathbb{R}$. For $a, b, c \in X$:

(M1) $p(a, b) = 0 \iff d_X(a, b) + d_Y(f(a), f(b)) = 0 \iff d_X(a, b) = 0$ and $d_Y(f(a), f(b)) = 0$ since $d_X(a, b) \geq 0$ and $d_Y(f(a), f(b)) \geq 0$. So $p(a, b) = 0 \iff a = b$ since $d_X(a, b) = 0 \iff a = b$ and $a = b$ implies $d_Y(f(a), f(b)) = 0$.

(M2) $p(a, b) = d_X(a, b) + d_Y(f(a), f(b)) = d_X(b, a) + d_Y(f(b), f(a)) = p(b, a)$ using the symmetry properties of d_X and d_Y .

(M3) $p(a, b) = d_X(a, b) + d_Y(f(a), f(b)) \leq d_X(a, c) + d_X(c, b) + d_Y(f(a), f(b)) + d_Y(f(b), f(c)) = p(a, c) + p(c, b)$ using the triangle inequality for d_X and d_Y .

- (b) Write down the definition of the diameter of a subset of a metric space.

Let A be a subset of a metric space (X, d) .

If $A = \emptyset$, $\text{diam } A = 0$.

If $\sup\{d(x, y) : x, y \in A\}$ exists, this is $\text{diam } A$.

If the supremum above does not exist, $\text{diam } A = \infty$ or is undefined.

(c) Now let

- $(X, d_X) = (\mathbb{R}, d_D)$
- $(Y, d_Y) = (\mathbb{R}, d_E)$
- $f(x) = x^2$

and define p as described above. In the metric space $(\mathbb{R}, p)$:

- Find all real numbers in the open ball $B(\sqrt{26}; 11)$. Show brief working.
- Find the diameter of the interval $[-4, 4]$. (No working required.)

$$B(\sqrt{26}; 11) = \{x \in \mathbb{R} : p(\sqrt{26}, x) < 11\} =$$

$$\{x \in \mathbb{R} : d_D(\sqrt{26}, x) + d_E(f(\sqrt{26}), f(x)) < 11\} =$$

$$\{x \in \mathbb{R} : x = \sqrt{26} \text{ or } 1 + |(\sqrt{26})^2 - x^2| < 11\} =$$

$$\{x \in \mathbb{R} : |x^2 - 26| < 10\} =$$

$$\{x \in \mathbb{R} : -10 < x^2 - 26 < 10\} =$$

$$\{x \in \mathbb{R} : 16 < x^2 < 36\} =$$

$$\{x \in \mathbb{R} : -6 < x < -4 \text{ or } 4 < x < 6\} = (-6, -4) \cup (4, 6).$$

Also $\text{diam } [-4, 4] = 17$.

$$[5+1+3+1=10]$$

2. Suppose that X and Y are metric spaces, and $f : X \rightarrow Y$ is a function.

(a) Prove that if f is continuous, then, for all $A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$.

Begin with $x \in \overline{A}$.

Then $x = \lim a_n$ for some sequence (a_n) in A .

By continuity of f , $f(a_n) \rightarrow f(x)$ as $n \rightarrow \infty$.

But $(f(a_n))$ is a sequence in $f(A)$, so $f(x) \in \overline{f(A)}$.

(b) Prove or disprove: f is continuous if and only if, for all $B \subseteq Y$, $\overline{f^{-1}(B)} = f^{-1}(\overline{B})$.

False; the forward implication does not hold:

Take $f : (\mathbb{R}, d_D) \rightarrow (\mathbb{R}, d_E)$ by $f(x) = x$ for all $x \in \mathbb{R}$.

This function is continuous, since all subsets of $(\mathbb{R}, d_D)$ are open, so inverse images of open sets under f will certainly be open.

However, take $B = (0, 1)$.

Then $\overline{f^{-1}(B)} = f^{-1}(B) = (0, 1)$.

But $f^{-1}(\overline{B}) = f^{-1}([0, 1]) = [0, 1]$.

[4+2=6]

3. (a) Prove that every convergent sequence in (X, d) is Cauchy, as used in line (3).

Take a sequence (x_n) such that $x_n \rightarrow x$ in X . Let $\epsilon > 0$ be given.

There exists $N \in \mathbb{N}$ such that $n \geq N$ implies that $d(x_n, x) < \epsilon/2$.

Then, for all $n, m \in \mathbb{N}$, $n \geq m \geq N$ implies that $d(x_n, x_m) < d(x_n, x) + d(x, x_m) < \epsilon/2 + \epsilon/2 = \epsilon$.

- (b) How do we know that $x_n \rightarrow a$ for some $a \in A$, as claimed in line (4)?

Since A is complete, every Cauchy sequence in A converges to a point of A .

- (c) Explain what is meant by the expression “limits are unique” (in line (5)) and prove it.

Statement: If $x_n \rightarrow x$ and $x_n \rightarrow y$ in a metric space, then $x = y$.

Proof: Suppose $x_n \rightarrow x$ and $x_n \rightarrow y$.

For all $n \in \mathbb{N}$, $0 \leq d(x, y) \leq d(x, x_n) + d(x_n, y)$.

As $n \rightarrow \infty$, $d(x, x_n) \rightarrow 0$ and $d(x_n, y) \rightarrow 0$.

So taking limits on both sides of the inequality gives $d(x, y) = 0$ and hence $x = y$.

- (d) Is the converse of this theorem true or false? Prove it or disprove it.

False:

Consider the metric space $X = ((0, 2), d_E)$ and subset $A = (0, 1]$.

Then A is a closed subset of X .

However, A is not complete, since $(\frac{1}{n})$ is a Cauchy sequence in A that does not converge to a point of A .

[2+2+3+2=9]

4. (a) i. Give brief reasons why, in any metric space, $B(a; r) \subseteq \text{int}B[a; r]$.
 ii. Give an instance where $B(a; r) \neq \text{int}B[a; r]$.

Clearly any open ball is a subset of the closed ball with the same centre and radius. An open ball in any metric space is an open set. Since the interior of a set is the largest open subset contained in the set, the result follows.

In $(\mathbb{R}, d_D)$, $B(5; 1) = \{5\}$ but $\text{int}B[5; 1] = B[5; 1] = \mathbb{R}$.

- (b) Prove that every compact metric space is bounded.

Suppose that X is a metric space that is not bounded.

Let $b \in X$.

Because X is not bounded, for every $n \in \mathbb{N}$, $X \neq B(b; n)$ so let $x_n \in X \setminus B(b; n)$.

Then $d(x_n, b) \geq n$ for all $n \in \mathbb{N}$.

Now (x_n) is a sequence without a convergent subsequence:

If $x_{n_k} \rightarrow x$ as $k \rightarrow \infty$, then we have, for all $k \in \mathbb{N}$:

$$n_k \leq d(x_{n_k}, b) \leq d(x_{n_k}, x) + d(x, b)$$

But $n_k \rightarrow \infty$ and $d(x_{n_k}, x) + d(x, b) \rightarrow d(x, b)$; a contradiction.

- (c) Prove or disprove: If (X, d_X) and (Y, d_Y) are connected metric spaces, and $X \times Y$ has a metric ρ that induces componentwise convergence, then $(X \times Y, \rho)$ is connected.

True.

Take $(a, b) \in X \times Y$.

Then $S = \{(x, b) : x \in X\}$ is topologically isomorphic to X , because ρ induces componentwise convergence. So S is connected.

And similarly, for all $x \in X$, $T_x = \{(x, y) : y \in Y\}$ is topologically isomorphic to Y . So T_x is connected.

Since $S \cap T_x \neq \emptyset$ for all $x \in X$, $S \cup (\cup T_x)$ is connected; but $S \cup (\cup T_x) = X \times Y$; so $X \times Y$ is connected.

[6+3+3=12]

5. (a) Given a metric space (X, d) , let $Y = X \times \mathbb{R}$ with any metric that induces componentwise convergence (the usual or the maximum or the taxi metric would do). Define $f : X \rightarrow Y$ by $f(x) = (x, 0)$. Then f is an isometry - the proof of this depends on what metric you put on Y . We check it with the usual metric ρ : $\rho(f(x), f(y)) = \rho((x, 0), (y, 0)) = \sqrt{(d(x, y))^2 + 0^2} = d(x, y)$ for all $x, y \in X$. Also, $f(X) = \{(x, 0) : x \in X\}$. We show $f(X)$ not open by showing $(f(X))'$ not closed: for any $x \in X$, take $(x, \frac{1}{n})$ in $(f(X))'$. Then $(x, \frac{1}{n}) \rightarrow (x, 0)$, which is in $f(X)$.
- (b) Prove or disprove: If (X, d) is a compact metric space and $f : (X, d) \rightarrow (X, d)$ is an isometry, then f is onto.

True.

Let $z \in X$.

Define (x_n) by choosing $x_1 = z$ and $x_{n+1} = f(x_n)$.

Since f is an isometry, $d(f(x_n), f(x_m)) = d(x_n, x_m)$ for all $n, m \in \mathbb{N}$.

So for $m > n$, $d(x_n, x_m) = d(f(x_n), f(x_m)) = d(x_{n+1}, x_{m+1})$.

Continuing like this gives, for $m > n$ and any $k \in \mathbb{N}$, $d(x_n, x_m) = d(x_{n+k}, x_{m+k})$.

Since X is compact, (x_n) has a convergent subsequence, which must be Cauchy.

Let $\epsilon > 0$ be given.

There exist $m > n$ such that $d(x_n, x_m) < \epsilon$.

Then $\text{dist}(z, f(X)) \leq d(z, x_{m-n+1}) = d(x_n, x_m) < \epsilon$.

So $\text{dist}(z, f(X)) = 0$.

Since X is compact, $f(X)$ is compact, so closed; so $z \in f(X)$.

[3]

THE END